

Data Description

You can find all data sets needed or generated during data processing for this project under this folder. Some information may be overlapped. To save the space, we don't upload images generated by code or the sub folders/files (sometimes CBGs are divided for parallel computing to save time).

1_data_for_extracting_information

We do not provide the raw weekly-patterns from SafeGraph since the data is not open to public anymore from their website.

1. all_info_before/after_aggregation.csv: this file shows detailed information of each store affiliated with Target and its competitors in NYC before/after aggregation. The aggregation is to combine the records of several stores in SafeGraph into one store (since we think they are actually the same store).
2. brand_info_nyc.csv: this file shows brand id for the stores used in this project.
3. cb_2019_us_bg_500k: this folder contains CBG information in US updated in 2019.
4. cbg_b01_nyc.csv: this file comes from SafeGraph where we get median income statistics for each CBG.
5. cbg_b02_nyc.csv: this file comes from SafeGraph where we get race statistics for each CBG.
6. cbg_b03_nyc.csv: this file comes from SafeGraph where we get race statistics especially in Hispanic for each CBG.
7. cbg_b15_nyc.csv: this file comes from SafeGraph where we get education statistics for each CBG.
8. cbg_b19_nyc.csv: this file comes from SafeGraph where we get household income statistics for each CBG.
9. cbg_b25_nyc.csv: this file comes from SafeGraph where we get housing units statistics for each CBG.
10. census_data_by_cbg.csv: population and median income for each CBG
11. core_poi-geometry_nyc.csv.: this file is used to extract the parking lot information of stores in NYC (if available).
12. first_step_data_with_competitors.csv: some information about stores included in our study.
13. NY_cbg_census.csv: some organized demographics data for NY too
14. nyc_cbgs.json: this file contains geography information of each CBG in NYC.
15. poi_count/diversity_per_cbg.csv: these files show the diversity and the number of POIs in each CBG. Diversity is computed based on the brand of each store.
16. poi_NYC_in_list.csv: NYC stores and their information extracted from core_poi-geometry.csv. We only put records of stores directly related to this project here.
17. SafeGraphPlacesGeoSupplementSquareFeet_202008_nyc.csv: this data set is used to pick the area of each store used in this project.
18. tl_2017_36_bg: this folder contains geometric information of NYC.

2_data_for_PSO_calibration

Here are 4 datasets that we created from all above provided datasets and unprovided weekly patterns in NYC from 2018~2021. We will use them in the large-scale computing process for research.

19. table_2018.csv: this table shows the features for each CBG in 2018.
20. table_2019.csv: this table shows the features for each CBG in 2019.

21. table_2020.csv: this table shows the features for each CBG in 2020.

22. table_2021.csv: this table shows the features for each CBG in 2021.

3_PSO_results

- **PSO_calibration_results**

- **norm_PSO_combination_2018**: under this folder, we have calibrated parameter values in 2018 of each normalization and PSO range combination.
- **norm_PSO_combination_2019**: under this folder, we have calibrated parameter values in 2019 of each normalization and PSO range combination.
- **norm_PSO_combination_2020**: under this folder, we have calibrated parameter values in 2020 of each normalization and PSO range combination.
- **norm_PSO_combination_2021**: under this folder, we have calibrated parameter values in 2021 of each normalization and PSO range combination.

- **simulated_annealing_data_2018_2021**

- norm_PSO_error_means_2018.csv: means of costs of each normalization and PSO range combination in 2018.
- norm_PSO_error_medians_2018.csv: medians of costs of each normalization and PSO range combination in 2018.
- norm_PSO_error_means_2019.csv: means of costs of each normalization and PSO range combination in 2019.
- norm_PSO_error_medians_2019.csv: medians of costs of each normalization and PSO range combination in 2019.
- norm_PSO_error_means_2020.csv: means of costs of each normalization and PSO range combination in 2020.
- norm_PSO_error_medians_2020.csv: medians of costs of each normalization and PSO range combination in 2020.
- norm_PSO_error_means_2021.csv: means of costs of each normalization and PSO range combination in 2021.
- norm_PSO_error_medians_2021.csv: medians of costs of each normalization and PSO range combination in 2021.

4_analysis

23. census_info_only_for_kmeans.csv: census info for CBGs that are used for Kmeans.

24. census_info_with_kmeans_label.cpg/dbf/shp/shx: kmeans clustering labels for CBGs.

25. did_general.csv: data ready for DID covering all CBGs.

26. did_sub_data: this folder contains tables of different sub groups that are used for DID.

27. kmeans_results.txt: Kmeans results saved from GeoDa.

28. parameter_values_ses_cluster_comparison_2019.csv: means of calibrated parameter values in 2019 in different communities.